

Monolingual Results

Language	Pair n-Gram Model (baseline 2020)	Encoder-decoder Transformer (baseline 2020)	Neural Transducer (baseline 2021/22)	Attentive LSTM (baseline 2024)
Adyghe	27	39	20	30
Arabic	43	43	53	45
Assamese	14	15	7	12
Belarusian	7	6	2	2
Bengali	70	78	68	67
Bulgarian	37	30	32	27
Burmese	39	41	29	35
Cebuano	22	26	20	20
Central Khmer	56	43	31	35
Classical Persian	51	58	57	51
Dutch	32	29	24	23
Eastern Armenian	17	16	15	20
Eastern Lawa	44	24	8	12
English	70	70	63	66
French	31	29	23	27
Georgian	0	4	0	1
German	49	56	46	45
Greek	27	26	20	27
Hindi	23	24	11	13
Hungarian	9	11	7	8
Icelandic	35	24	12	16
Indonesian	58	52	64	73
Iranian Persian	66	63	65	65
Irish	57	46	43	39
Italian	22	21	15	16
Japanese	23	20	10	12
Korean	81	89	23	100
Latvian	51	51	51	50
Lithuanian	32	35	33	31
Macedonian	6	5	5	5
Maltese	27	24	17	22
Norwegian Nynorsk	61	69	66	69
Pashto	70	68	67	66
Polish	9	8	4	7
Romanian	10	17	9	17
Russian	31	21	23	25
Serbo-Croatian	84	69	64	69
Shan	6	10	5	4
Slovenian	73	52	56	50
Spanish	3	10	4	3
Swedish	67	68	59	61
Tagalog	11	17	13	11

Language	Pair n-Gram Model (baseline 2020)	Encoder-decoder Transformer (baseline 2020)	Neural Transducer (baseline 2021/22)	Attentive LSTM (baseline 2024)
Thai	70	49	39	42
Ukrainian	27	26	19	18
Urdu	67	66	72	77
Uyghur	0	3	0	2
Vietnamese	44	20	5	4
Welsh	33	20	12	19
Macro-average WER	37.33	35.23	28.98	32.06

Error Analysis

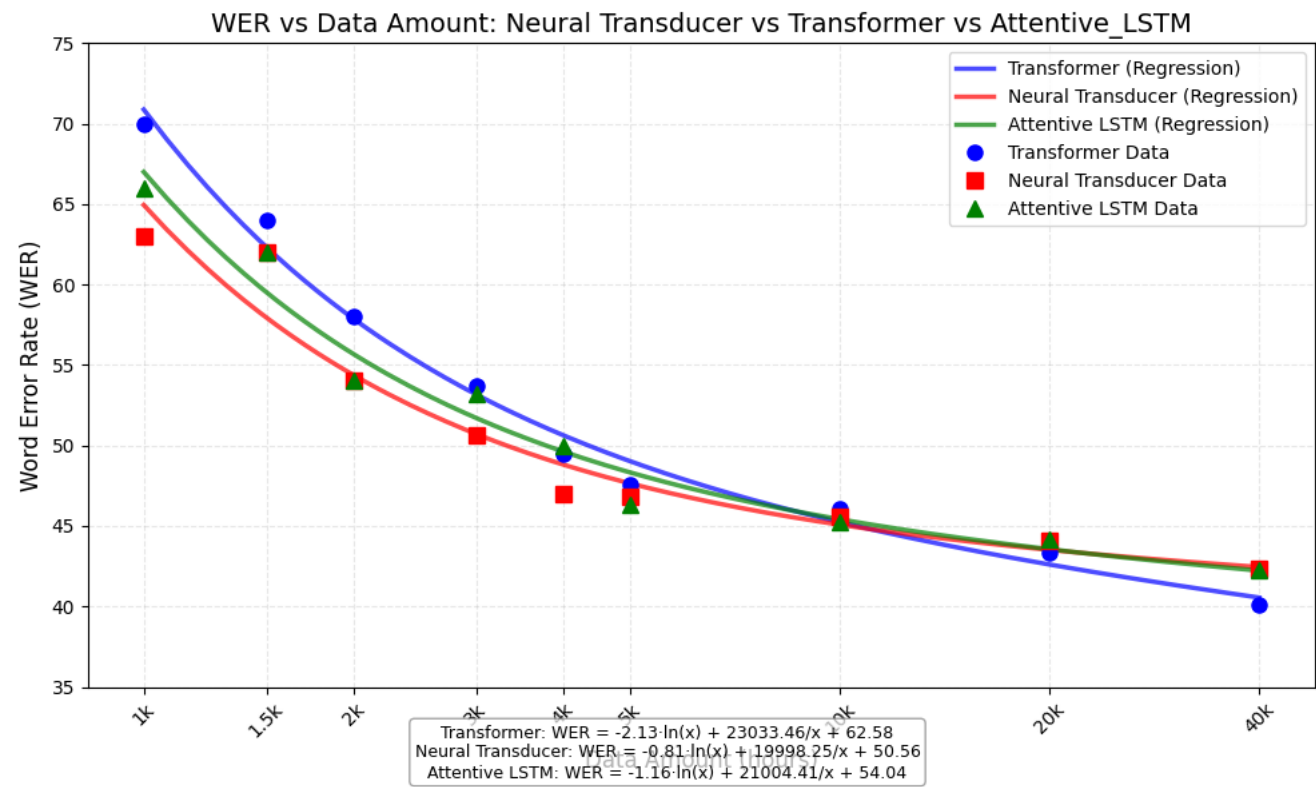
Korean

	Neutral Transducer	Encoder-Decoder Transformer	Attentive LSTM
Hangul	23	89	100
Jamo	23	28	28

How Data Amount Affects Performance of Transformer Model

Comparing best performing model and transformer on different amount fo English data

Data Amount	Neutral Transducer	Encoder-Decoder Transformer	Attentive LSTM
1k	63.00	70.00	66.00
1.5k	62.00	64.00	62.00
2k	54.00	58.00	54.00
3k	50.67	53.67	53.20
4k	47.00	49.50	50.00
5k	46.80	47.60	46.30
10k	45.60	46.10	45.21
20k	44.05	43.35	44.13
40k	42.38	40.10	42.23



Multilingual Approach

A tag comprising a language code (e.g. <[ENG]> for English, <[DEU]> for German) is prepended to each grapheme sequence source. The datasets of each language are simply concatenated.

Does converting the language tag to the corresponding script of the language improve the model performance?

Using `slavic_cyrillic` dataset we got following validation WER:

Tag in	Neural Transducer	Encode-Decoder Transformer	Attentive LSTM
Uppercase Cyrillic	39.67	28.00	28.00
Uppercase Latin	31.00	28.00	27.00
Lowercase Latin	32.33	28.00	27.00
No tag	45.00	44.67	44.67

Results

Same Language Family and Same Script

			Transducer			Transformer			Attentive LSTM			ByT5
Family	Script	Language	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ	
Indo-European (Germanic)	Latin	German	45	46	-1	50	56	-6	45	45	0	51
		Swedish	60	59	1	59	68	-9	61	61	0	64
		Dutch	28	24	4	20	29	-9	21	23	-2	22
Average					1.33			-8.00			-0.67	
Indo-European (Romance)	Latin	Italian	30	15	15	21	21	0	17	16	1	17
		Spanish	11	4	7	4	10	-6	3	3	0	3
		Romanian	16	9	7	17	17	0	9	17	-8	14
Average					9.67			-2.00			-2.33	
Indo-European (Slavic)	Cyrillic	Russian	27	23	4	18	21	-3	22	25	-3	24
		Ukrainian	22	19	3	27	26	1	19	18	1	35
		Bulgarian	35	32	3	27	30	-3	31	27	4	27

				Transducer		Transformer		Attentive LSTM			ByT5	
Average				3.33		-1.67		0.67				
Indo-European (Slavic)	Latin	Serbo-Croatian	67	64	3	63	69	-6	68	69	-1	62
		Polish	13	4	9	9	8	1	9	7	2	8
		Slovenian	59	56	3	52	52	0	50	50	0	53
Average				5.00		-1.67		0.33				
Indo-European (Indo-Iranian)	Arabic	Pashto	80	67	13	67	68	-1	68	66	2	70
		Classical Persian	63	57	6	53	58	-5	59	51	8	45
		Urdu	77	72	5	67	66	1	69	77	-8	62
Average				8.00		-1.67		0.67				
Austronesian	Latin	Indonesian	66	64	2	57	52	5	55	73	-18	51
		Cebuano	26	13	13	24	26	-2	22	20	2	26
		Tagalog	15	20	-5	18	17	1	16	11	5	10
Average				3.33		1.33		-3.67				

Different Language Family and Same Script

			Transducer			Transformer			Attentive LSTM			ByT5
Family	Script	Language	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ	
Afro-Asiatic	Arabic	Arabic	54	53	1	38	43	-5	40	45	-5	42
Indo-European		Classical Persian	58	57	1	42	58	-16	45	51	-6	45
Turkic		Uyghur	10	0	10	3	3	0	2	2	0	0
Average			4.00			-7.00			-3.67			
Indo-European	Latin	Italian	18	15	3	25	21	4	25	16	9	17
Austronesian		Indonesian	58	64	-6	52	52	0	53	73	-20	51
Uralic		Hungarian	8	7	1	10	11	-1	6	8	-2	10
Average			-0.67			1.00			-4.33			

Larger Model/Dataset

		Transducer				Transformer			Attentive LSTM			ByT5
Family	Script	Language	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ	
Indo-European		Italian	16	15	1	22	21	1	19	16	3	17
Austronesian		Indonesian	56	64	-8	52	52	0	57	73	-16	51
Uralic		Hungarian	9	7	2	12	11	1	7	8	-1	10
Afro-Asiatic		Maltese	19	17	2	18	24	-6	17	22	-5	19
Average					-0.75	-1.00			-4.75			

Same Language Family and Different Script

			Transducer			Transformer			Attentive LSTM			ByT5
Family	Script	Language	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ	
Indo-European	Arabic	Classical Persian	63	57	6	40	58	-18	44	51	-7	45
	Cyrillic	Russian	46	23	23	21	21	0	24	25	-1	24
	Latin	Italian	20	15	5	22	21	1	18	16	2	17
Average					11.33			-5.67			-2.00	

Larger Model/Dataset

		Transducer			Transformer			Attentive LSTM			ByT5
Family	Script	Language	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ

			Transducer			Transformer			Attentive LSTM			ByT5
Indo-European	Arabic	Classical Persian	50	57	-7	43	58	-15	44	51	-7	45
	Cyrillic	Russian	34	23	11	26	21	5	26	25	1	24
	Latin	Italian	23	15	8	24	21	3	18	16	2	17
	Armenian	Eastern Armenian	19	15	4	16	16	0	18	20	-2	25
Average			4.00			-1.75			-1.50			

Effect of Transliteration

Latin transliteration of languages with Cyrillic script

Monolingual

			Transducer			Transformer			Attentive LSTM		
	Latin	Cyrillic	Δ	Latin	Cyrillic	Δ	Latin	Cyrillic	Δ		
Russian	20	23	-3	22	21	1	25	30	-5		
Ukrainian	22	19	3	24	26	-2	18	20	-2		
Bulgarian	34	32	2	35	30	5	27	29	-2		
Macedonian	6	5	1	5	5	0	5	5	0		
Average	0.75			1.00			-2.25				

Multilingual Dataset - Slavic Latin

			Transducer			Transformer			Attentive LSTM			ByT5
Language	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ			
Bulgarian*	31	34	-3	27	35	-8	23	27	-4			28
Russian*	19	20	-1	24	22	2	20	25	-5			31
Ukrainian*	23	22	1	18	24	-6	19	18	1			34
Serbo-Croatian	76	64	12	63	69	-6	61	69	-8			62
Polish	13	4	9	7	8	-1	8	7	1			8
Slovenian	54	56	-2	57	52	5	49	50	-1			53
Macedonian*	4	6	-2	3	5	-2	4	5	-1			6
Average	2.00			-2.29			-2.43					

Latin transliteration of languages with Arabic script

Monolingual

	Transducer				Transformer				Attentive LSTM			
	uroman	epitran	Arabic	Δ	uroman	epitran	Arabic	Δ	uroman	epitran	Arabic	Δ
Arabic	70	63	53	17	67		43	24	70		45	25
Classical Persian	48	58	57	-9	49		58	-9	50		51	-1
Pashto	74	/	67	7	76	/	68	8	72	/	66	6
Urdu	72	73	72	0	67		66	1	78		77	1
Uyghur	28	4	0	28	38		3	35	35		2	33
Average				8.6				11.8				12.8

Multilingual(uroman)

			Transducer			Transformer			Attentive LSTM		
	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ		
Arabic*	76	70	6	42	67	-25	65	70	-5		
Classical Persian*	66	48	14	42	49	-7	50	50	0		
Urdu*	82	72	10	64	67	-3	70	78	-8		

	Transducer			Transformer			Attentive LSTM		
Uyghur*	37	28	9	36	38	-2	33	35	-2
Average			9.4			-8.6			-2.6

Multilingual(epitran)

	Transducer			Transformer			Attentive LSTM		
	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ
Arabic*	58	63	-5						
Classical Persian*	66	58	8						
Urdu*	77	73	4						
Uyghur*	10	4	6						
Average			3.3						

Latin transliteration of Indo-European languages

Monolingual

	Transducer			Transformer			Attentive LSTM		
	Latin	Armenian	Δ	Latin	Armenian	Δ	Latin	Armenian	Δ
Eastern Armenian	40	15	25	43	16	27	39	20	19
	Latin	Greek	Δ	Latin	Greek	Δ	Latin	Greek	Δ
Greek	30	20	10	35	26	9	35	27	8

Multilingual

	Transducer			Transformer			Attentive LSTM		
	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ
Classical Persian*	58	48	10	50	49	1	46	50	-4
Russian*	39	20	19	29	22	7	30	27	3
Italian	21	15	6	19	21	-2	15	16	-1
Eastern Armenian*	45	40	5	40	43	-3	42	39	3
German	47	46	1	45	56	-11	51	45	6
Greek*	32	30	2	33	35	-2	33	35	-2
Irish	45	43	2	43	46	-3	41	39	2
Average			6.43			-1.86			1.00

Influence of Languages on each other in a dataset

Language Family and Script

	Italian			Classical Persian		
	1F1S	DF1S	1FDS	1F1S	DF1S	1FDS
Transducer	30	18	20	63	58	63
Transformer	21	25	22	53	42	40
Attentive LSTM	17	25	18	59	45	44
	Indonesian			Russian		
Transducer	66	58		27		46
Transformer	57	52		18		21
Attentive LSTM	55	53		22		24

Classical/Iranian Persian and Uyghur

	Transducer	Transformer	Attentive LSTM
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Language	Transducer			Transformer			Attentive LSTM		
	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ
Arabic	54	53	1	38	43	-5	40	45	-5
Classical Persian	58	57	1	42	58	-16	45	51	-6
Uyghur	10	0	10	3	3	0	2	2	0
Arabic	68	53	15	39	43	-4	43	45	-2
Iranian Persian	68	65	3	61	63	-2	63	65	-2
Uyghur	11	0	11	3	3	0	2	2	0

Maltese

Language	Transducer			Transformer			Attentive LSTM		
	Multi	Mono	Δ	Multi	Mono	Δ	Multi	Mono	Δ
Italian	28	15	13	23	21	2	17	16	1
Maltese	30	17	13	16	24	-8	19	22	-3
Arabic*	77	70	7	47	67	-20	50	70	-20
Average			11.00			-8.67			-7.33

Reference Result from previous shared task

2022

Model

Baseline: A neural transducer system using an imitation learning paradigm (dyNET framework)

Submissions:

- 1. [Tü-G2P](#): A series of sequence labelling systems to G2P tasks, which use simpler alignment rather than dynamic transducer-based alignment.(Pytorch)
- 2. [Hammond \(Repo\)](#): A non-neural system based on OpenFST and uses weighted finite-state transducers and expectation-maximization to compute the best many-to-many alignment of letters and phonetic symbol
- 3. ~~[mSLAM](#): Non-archival, abstract only, useless~~
- 4. ~~[NFST](#): Non-archival, abstract only, useless~~

Reference Result:

- Baseline

Language	Bengali	Burmese	German	Irish	Italian	Persian	Swedish	Tagalog	Thai	Ukrainian	Macro-average
WER	67.12	29.00	42.00	38.00	15.00	59.65	45.00	20.00	21.00	32.00	36.88

- Hammond (trigram alignment)

Language	Bengali	Burmese	German	Irish	Italian	Persian	Swedish	Tagalog	Thai	Ukrainian	Macro-average
WER	68.49	48.00	61.00	51.00	25.00	67.86	55.00	18.00	72.00	50.00	51.63

2024

Model

Baseline: attentive_gru, attentive_lstm, gru, hard_attention_gru, lstm, pointer_generator_gru, transducer_gru, transformer(20\40\60 epochs)

Reference Result:

- Baseline

Models	gru	lstm	attentive_gru	attentive_lstm	hard_attention_gru	hard_attention_gru (Arab)	pointer_generator_gru	transducer_gru	transformer
WER (%)	43.75	44.25	63.08	47.25	40.67	31.33	62.17	69.33	78.25

- Best performance model on all datasets(hard_attenton_gru)

Languages	Arabic	Bulgarian	English	Persian	Indonesian	Macedonian	Pashto	Russian	Spanish	Tagalog	Ukrainian	Urdu
WER	31.33	20.00	58.00	29.67	55.33	3.67	44.33	10.33	5.00	40.33	15.67	64.00

- GRU、LSTM、Transformer on different languages

Model/Languages	English	Pashto	Russian	Spanish
GRU	31.00	39.00	14.00	9.00
LSTM	48.33	57.67	10.33	9.00
Transformer	81.33	77.00	35.67	24.67

- Comparison between hard_attention_gru、attentive_lstm

Model / Language	Korean	Bengali	Indonesian	Pashto	Swedish
hard_attention_gru	98.00	87.00	85.00	78.00	66.00
attentive_lstm	100.00	67.00	73.00	66.00	61.00

- Comparison between attentive_gru and attentive_lstm

The comparison of validation accuracy (val_accuracy) between attentive_gru (GRU) and attentive_lstm (LSTM) across all languages shows:

- LSTM val_accuracy higher than GRU: 20 times
- LSTM val_accuracy lower than GRU: 22 times
- LSTM val_accuracy equal to GRU: 6 times
- Average LSTM val_accuracy: 0.6987
- Average GRU val_accuracy: 0.7049

This indicates that, on average, attentive_gru achieves slightly higher validation accuracy than attentive_lstm, although the performance varies across different languages.

- attentive_lstm performance on Adyghe and Bengali with different parameters(DEV_WER)

Params	Adyghe WER	Bengali WER	Dutch WER	Urdu WER
default	31.00	65.00	23.00	77.00
4 encoder_layers + 1 decoder_layer	51.00	70.00	45.00	80.00
256 embedding_size + 1024 hidden_size	37.00	67.00	30.00	78.00
0.1 label_smoothing	33.00	65.00	21.00	75.00

- Summary of the G2P performance of the Transphone toolkit on several languages

Language	Dutch	English	French	Italian	Spanish
WER	51.00	76.00	7.00	21.00	16.00